

I. Optimal Tree Construction for Regression (1–10)

1. **Q:** What is the name of the algorithm used to determine the architecture of a decision tree?

A: Recursive binary splitting.

2. **Q:** What does recursive binary splitting iterate over to find the best split at any node?

A: It iterates over **all features and all possible split points**.

3. **Q:** The optimal split of a region (R) is defined by finding the optimal combination of which two variables?

A: The **feature (j)** and the **split point (s)**.

4. **Q:** Define the first region ($R_1(j, s)$) after a split.

A:

[
 $R_1(j, s) = \{\vec{x} \mid x_j \leq s\}$
]

5. **Q:** Define the second region ($R_2(j, s)$) after a split.

A:

[
 $R_2(j, s) = \{\vec{x} \mid x_j > s\}$
]

6. **Q:** For regression, what loss function evaluates the split quality?

A: **Sum of Squared Errors (SSE)**.

7. **Q:** The SSE formula sums over which two items when evaluating a split?

A: The **SSE of region (R_1)** and the **SSE of region (R_2)**.

8. **Q:** The optimal constant value ($\bar{y}_{R_1}$) is calculated as what property of true values?

A: The **mean of (y_i) values in (R_1)**.

9. **Q:** Formally, what is $\sum_{\vec{x}_i \in R_1} (y_i - \bar{y}_{R_1})^2$ called?

A: The **Sum of Squared Errors (SSE) for region (R_1)**.

10. **Q:** The greedy approach ensures that at each step, the algorithm chooses the split that results in the greatest reduction in what metric?

A: The **overall SSE** (or total variance) of the tree.

II. Tree Pruning and Regularization (11–20)

11. Q: What is the main problem of growing a tree until every leaf contains one data point?

A: **Overfitting** — the tree may not generalize to new data.

12. Q: What process is applied to overly large trees to improve generalization?

A: **Pruning**.

13. Q: Symbolic representation of the cost-complexity measure?

A: $(C_{\alpha}(T))$

14. Q: What are the two terms in $(C_{\alpha}(T))$?

A: **SSE of the tree + penalty for tree complexity** $(\alpha \cdot |T|)$

15. Q: What does $\sum_{k=1}^{|T|} \sum_{\vec{x}_i \in R_k} (y_i - \hat{c}_k)^2$ represent?

A: The **total SSE across all leaves** of the tree.

16. Q: What does $(|T|)$ represent in $(C_{\alpha}(T))$?

A: The **number of terminal nodes (leaves)** in the tree.

17. Q: What is the hyperparameter (α) called?

A: The **complexity parameter** or **regularization parameter**.

18. Q: How is the base tree (T_0) obtained?

A: By **growing a full tree** without pruning.

19. Q: What technique determines the optimal (α) ?

A: **Cross-validation**.

20. Q: What happens to the penalty term $(\alpha \cdot |T|)$ as tree complexity $(|T|)$ increases?

A: The penalty **increases**, discouraging overly complex trees.

III. Classification Decision Trees (21–30)

21. Q: For binary classification, what is an alternative name for the Log Loss function?

A: **Binary Cross-Entropy Loss**.

22. Q: Formula for Log Loss for binary classification?

A:

$$L_{\log} = -\frac{1}{n} \sum_{i=1}^n \big[y_i \log(\hat{p}_i) + (1-y_i) \log(1-\hat{p}_i) \big]$$

23. Q: In the Log Loss formula, what does $(\hat{p}_i)$ represent?

A: The **predicted probability** of class 1 for datapoint (i) .

24. Q: The term $(y_i \log(\hat{p}_i))$ contributes only when (y_i) equals what?

A: 1 (the positive class).

25. Q: For multiclass classification, what is the commonly used loss function?

A: **Multinomial cross-entropy (Softmax Loss).**

26. Q: Formula for multinomial cross-entropy loss with (C) classes?

A:

$$L = -\sum_{i=1}^n \sum_{c=1}^C I(y_i = c) \log(\hat{p}_{ic})$$

27. Q: In multinomial loss, what does $I(y_i = c)$ represent?

A: An **indicator function**: 1 if true class is (c) , else 0.

28. Q: In a classification tree, what does the leaf prediction $(\hat{P}_{ic})$ represent?

A: The **probability of datapoint (i) belonging to class (c) .**

29. Q: For classification, what are two other splitting criteria besides Log Loss?

A: **Gini Index** and **Entropy**.

30. Q: When a node is split, maximizing homogeneity corresponds to what property in the loss function?

A: **Minimizing impurity** (Gini Index or Entropy) in child nodes.